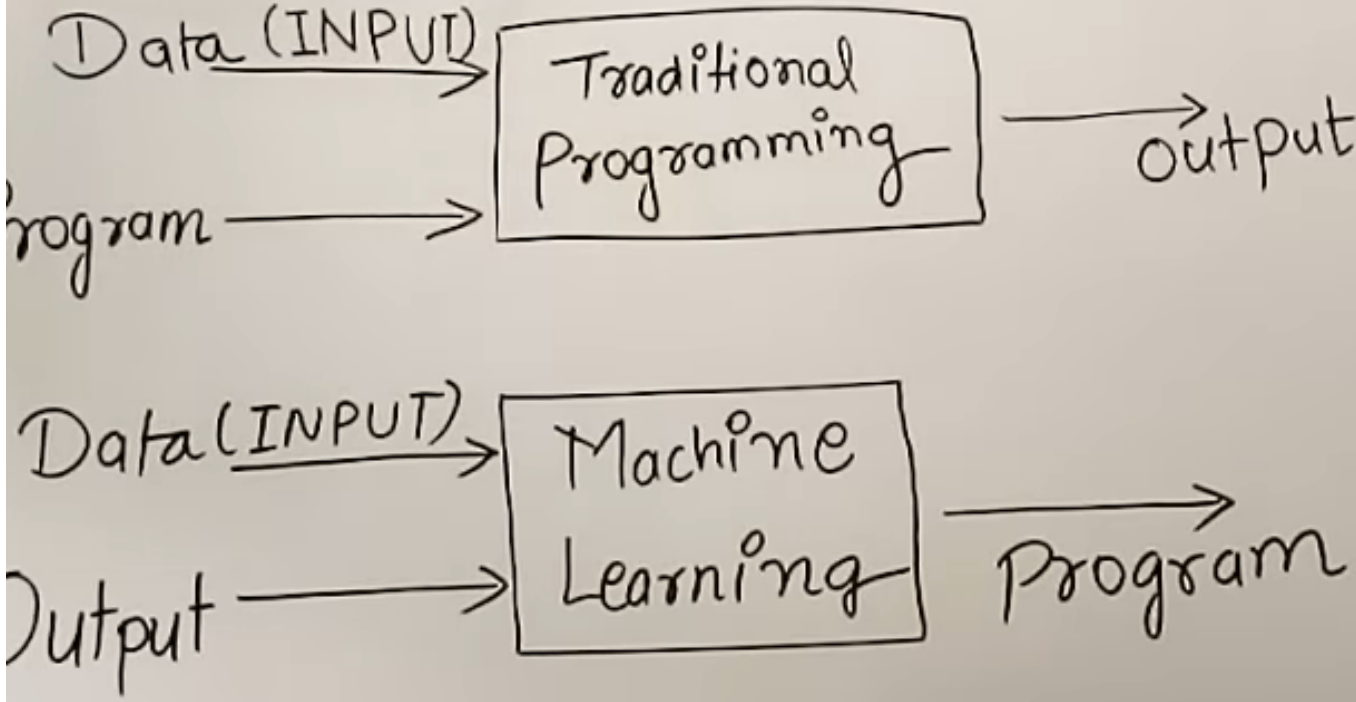
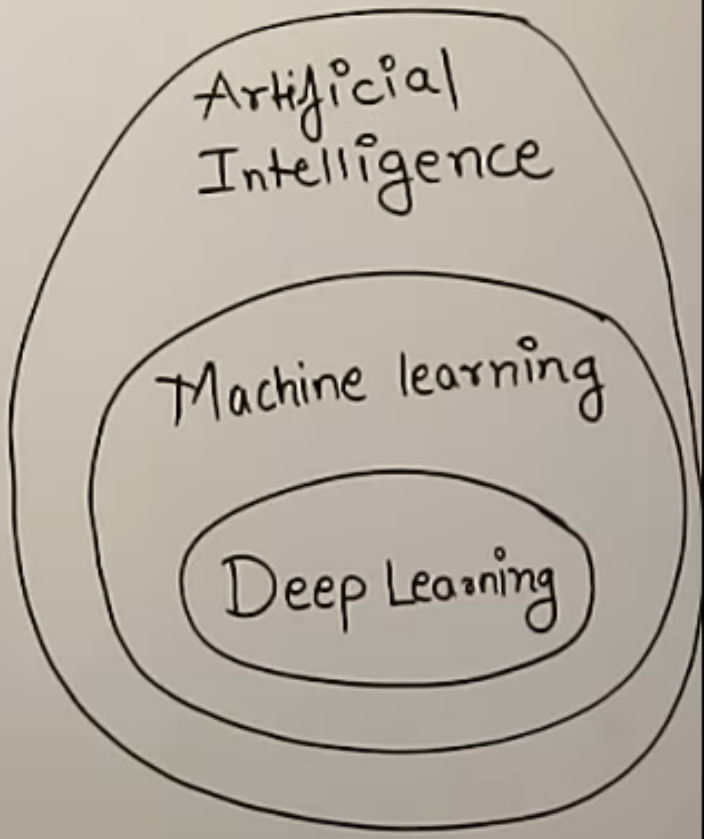
Machine learning

- It is the field of study that gives Computers the Capability to learn without being explicitly programmed.





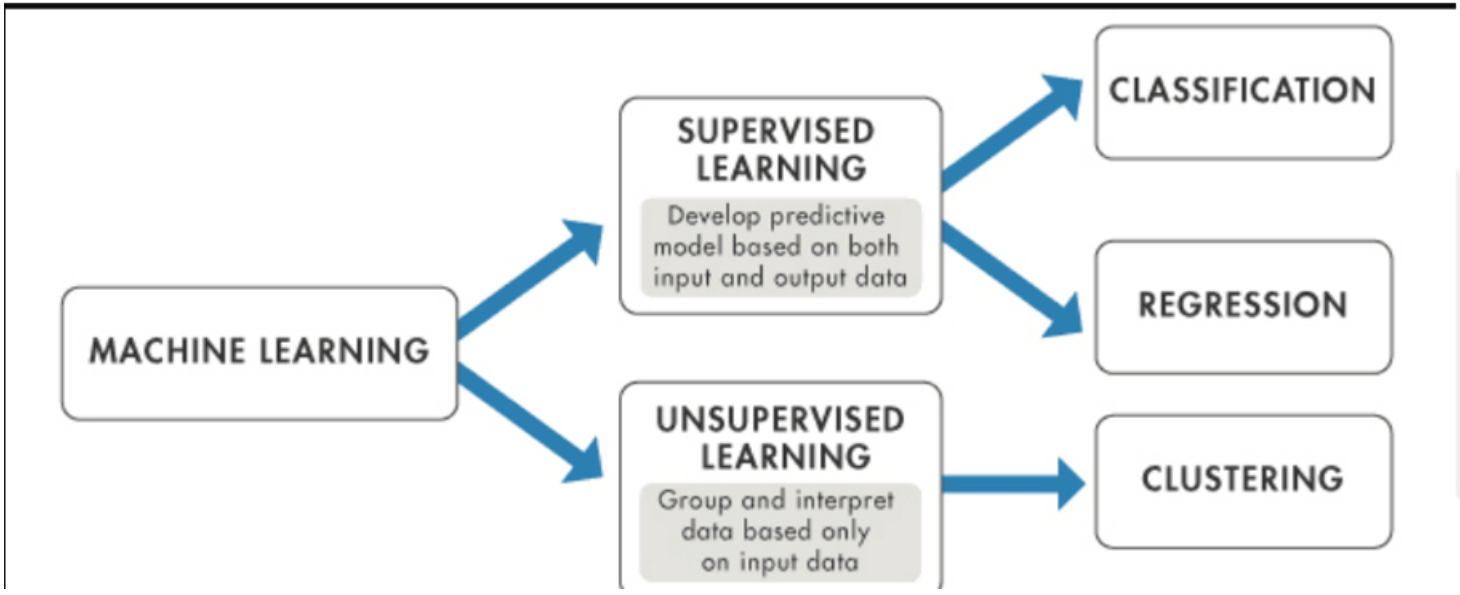
Definitions

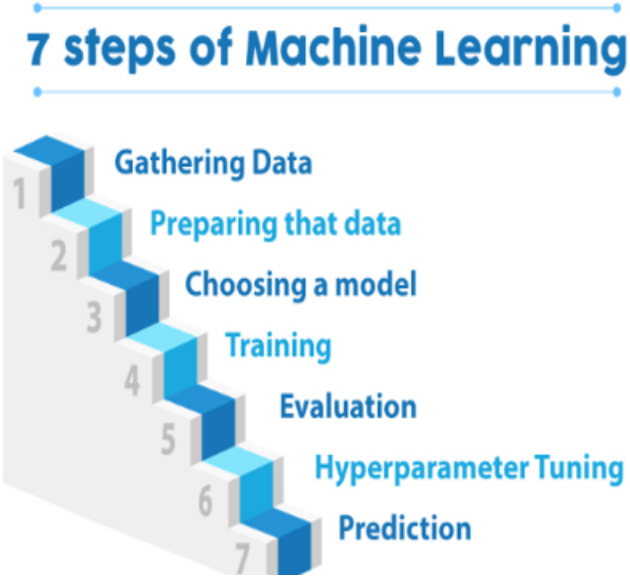
Artificial intelligence (AI): A broad discipline with the goal of creating intelligent machines, as opposed to the natural intelligence that is demonstrated by humans and animals. It has become a somewhat catch all term that nonetheless captures the long term ambition of the field to build machines that emulate and then exceed the full range of human cognition.

Machine learning (ML): A subset of AI that often uses statistical techniques to give machines the ability to "learn" from data without being explicitly given the instructions for how to do so. This process is known as "training" a "model" using a learning "algorithm" that progressively improves model performance on a specific task.

Reinforcement learning (RL): An area of ML concerned with developing software agents that learn goal-oriented behavior by trial and error in an environment that provides rewards or penalties in response to the agent's actions (called a "policy") towards achieving that goal.

Deep learning (DL): An area of ML that attempts to mimic the activity in layers of neurons in the brain to learn how to recognise complex patterns in data. The "deep" in deep learning refers to the large number of layers of neurons in contemporary ML models that help to learn rich representations of data to achieve better performance gains.



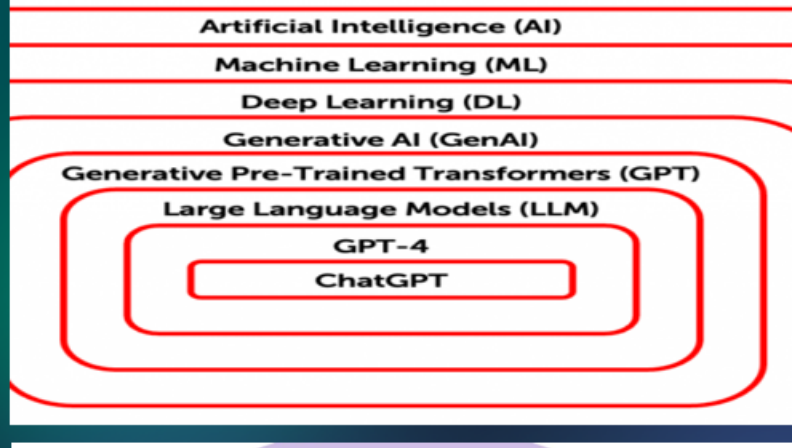


What are Machine learning algorithms?

Machine learning algorithms are the engines of machine learning, meaning it is the **algorithms that turn a data set into a model.**



AI Terminology



A model for thinking about AI, ML, DL, and generative AI



Artificial intelligence (AI)

Any technique that allows computers to mimic human intelligence using logic, if-then statements, and machine learning



Machine learning (ML)

A subset of AI that uses machines to search for patterns in data to build logic models automatically



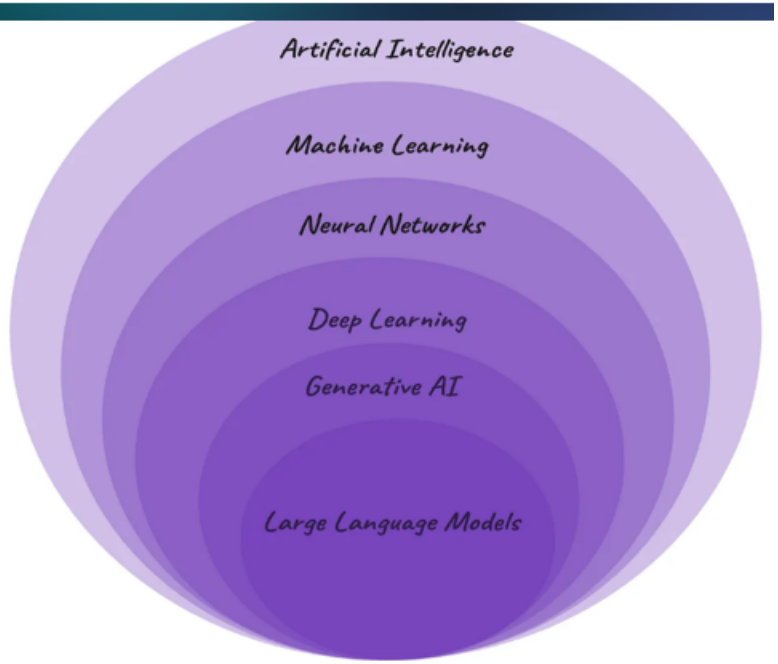
Deep learning (DL)

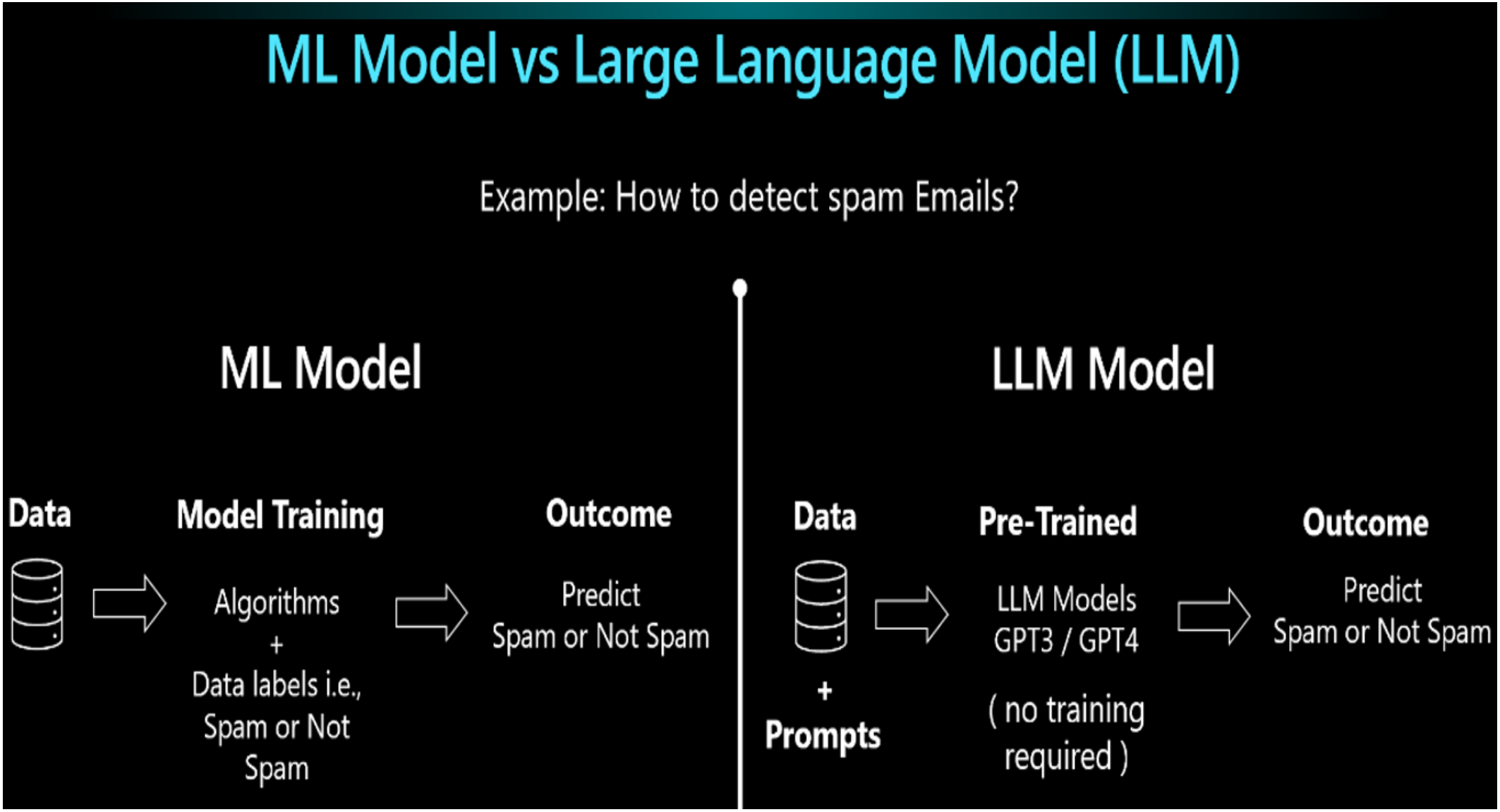
A subset of ML composed of deeply multi-layered neural networks that perform tasks like speech and image recognition



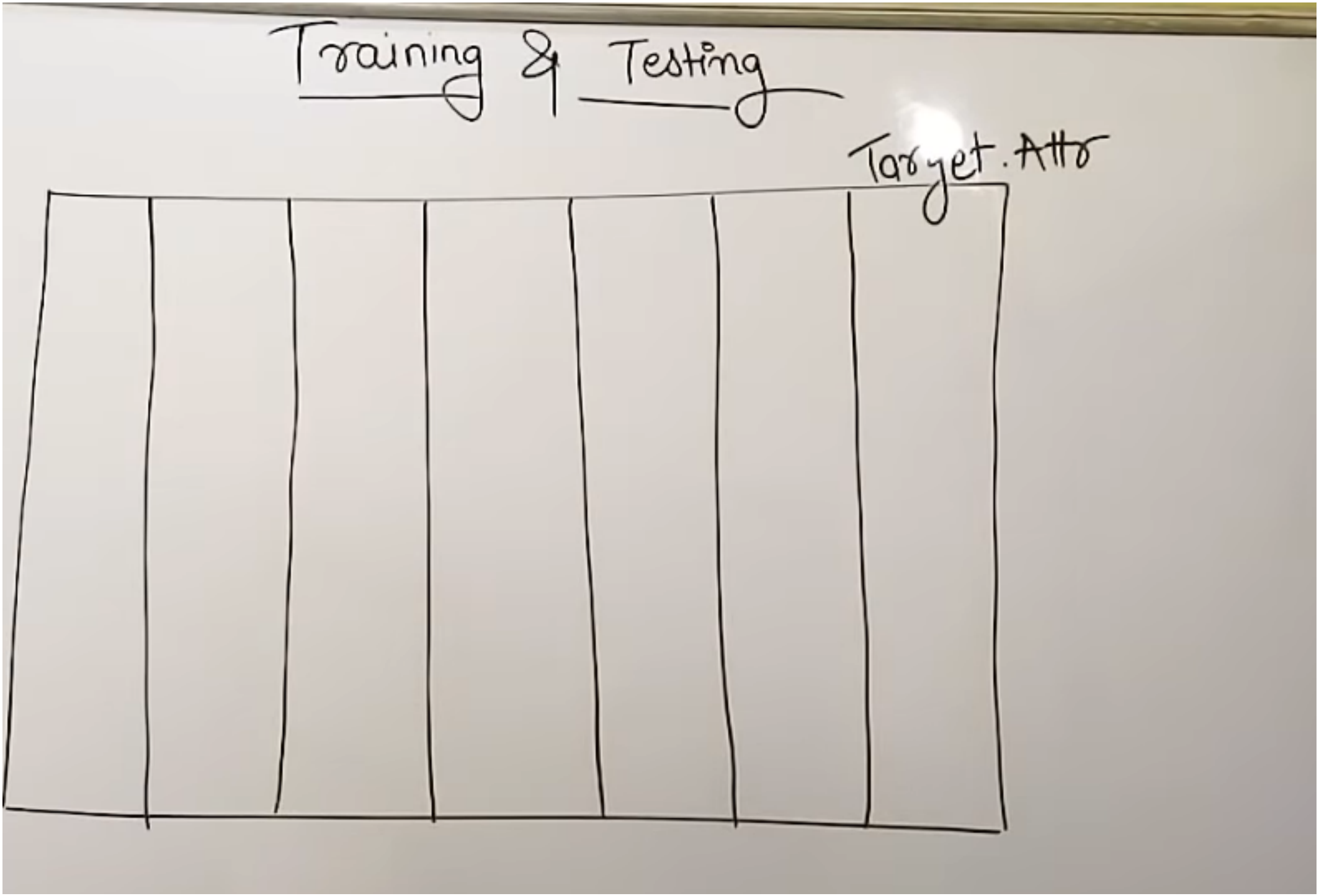
Generative AI

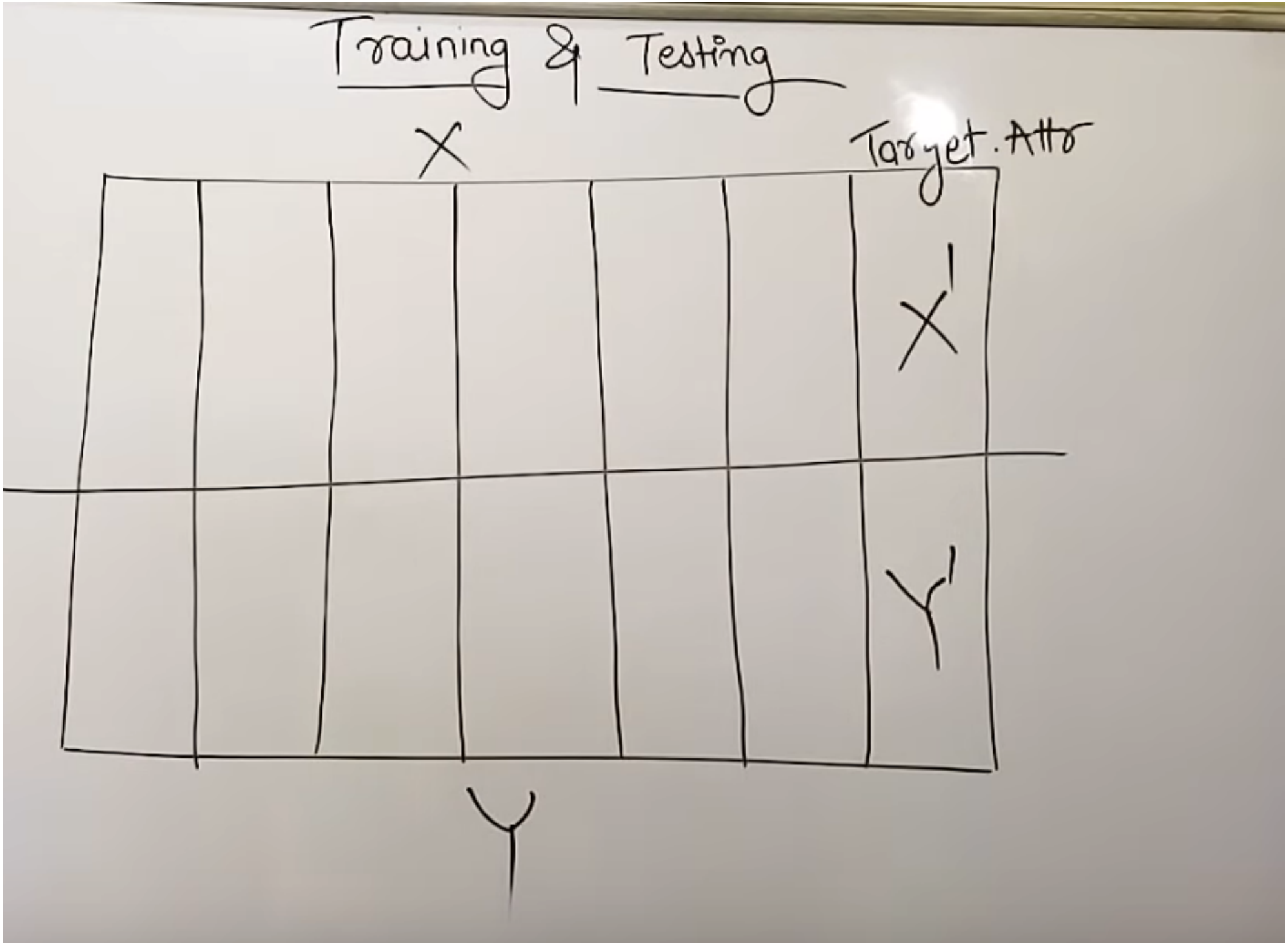
Powered by large models that are pretrained on vast corpora of data and commonly referred to as foundation models (FMs)

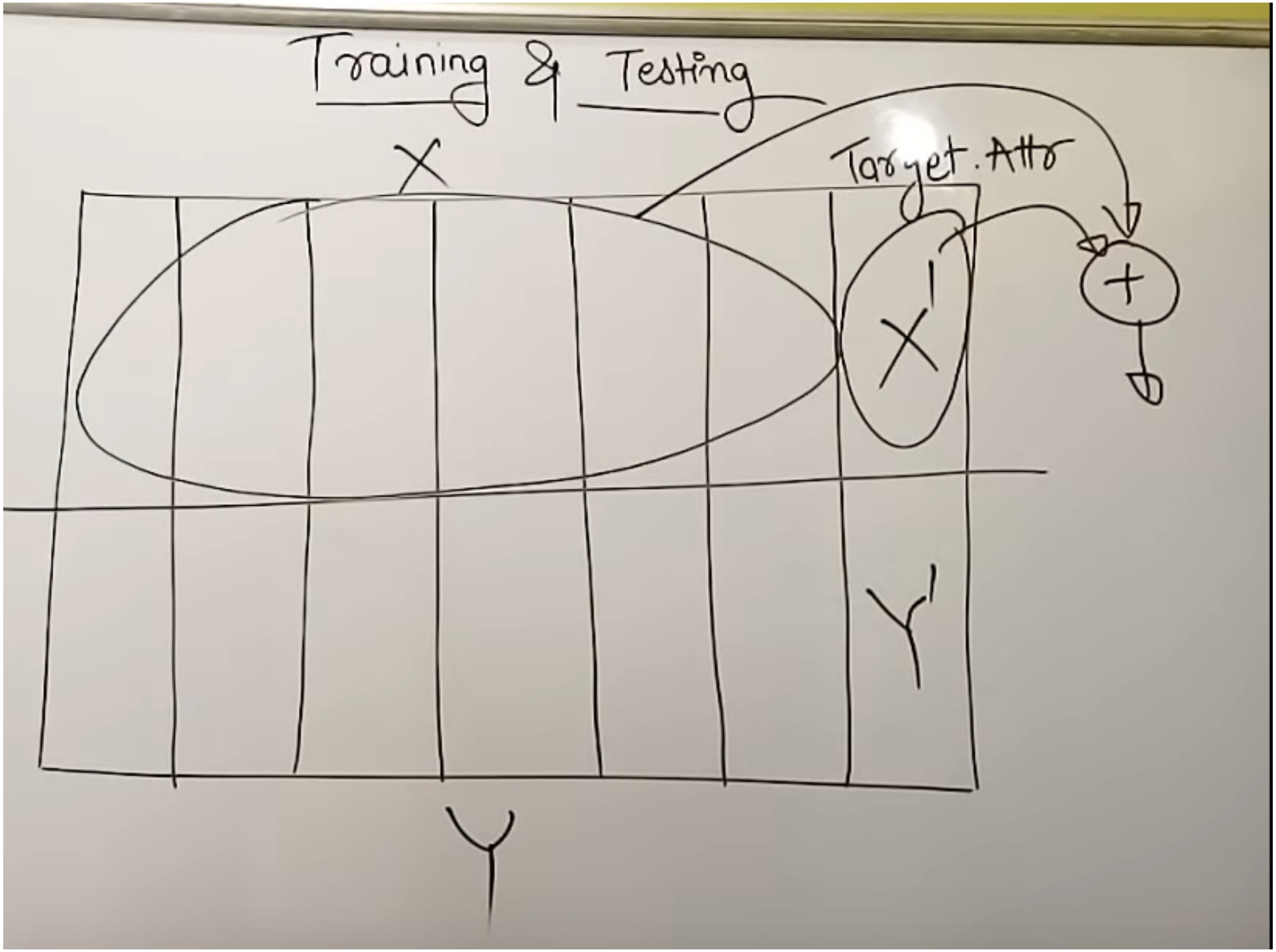


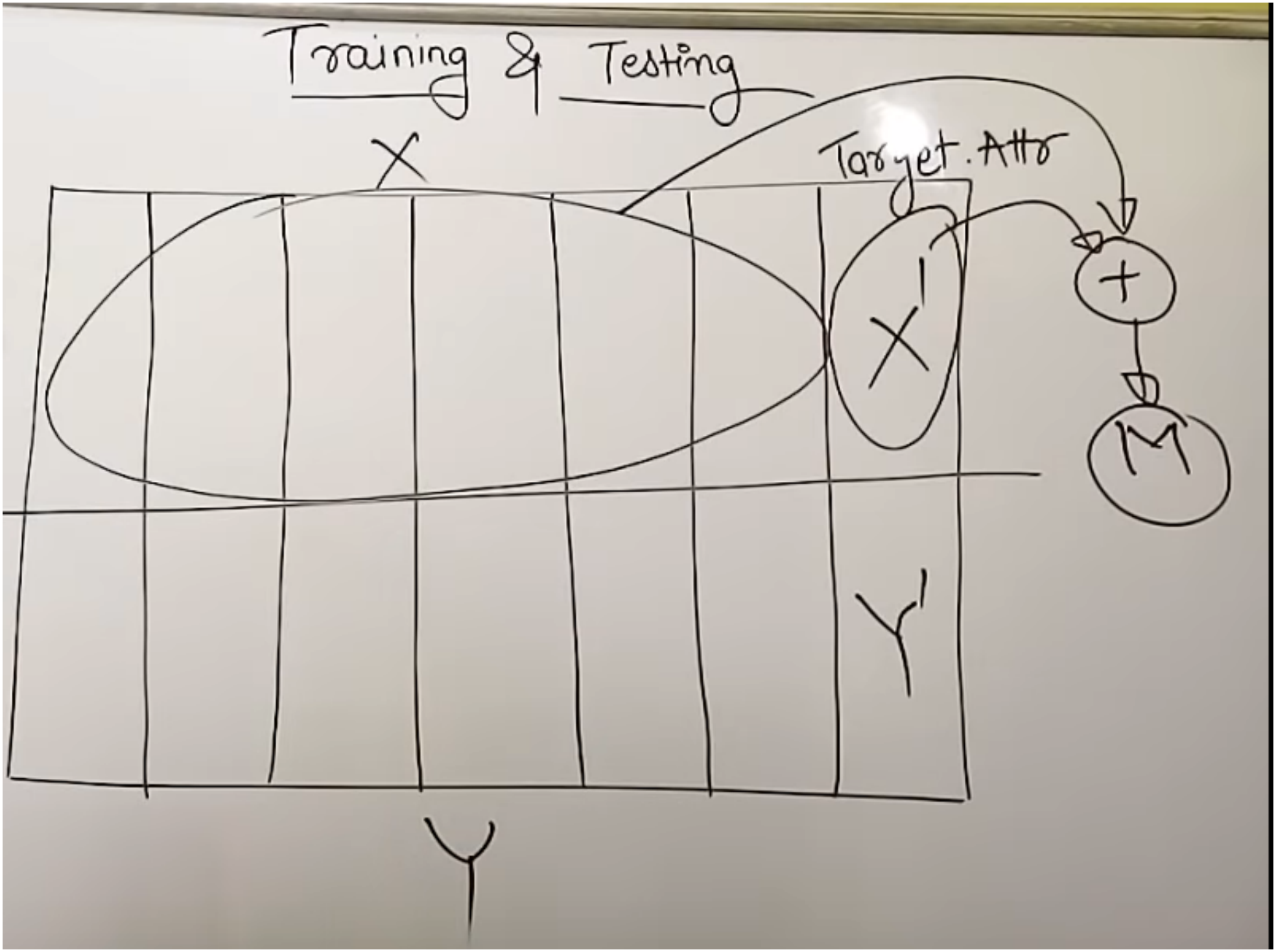

ML Model vs Large Language Model (LLM)
Example: How to detect spam Emails?
ML Model
Data
Model Training
Algorithms
+
Data labels i.e.,
Spam or Not
Spam
Outcome
Predict
Spam or Not Spam
LLM Model
Data
+
Prompts
Pre-Trained
LLM Models
GPT3 / GPT4
(no training
required)
Outcome
Predict
Spam or Not Spam

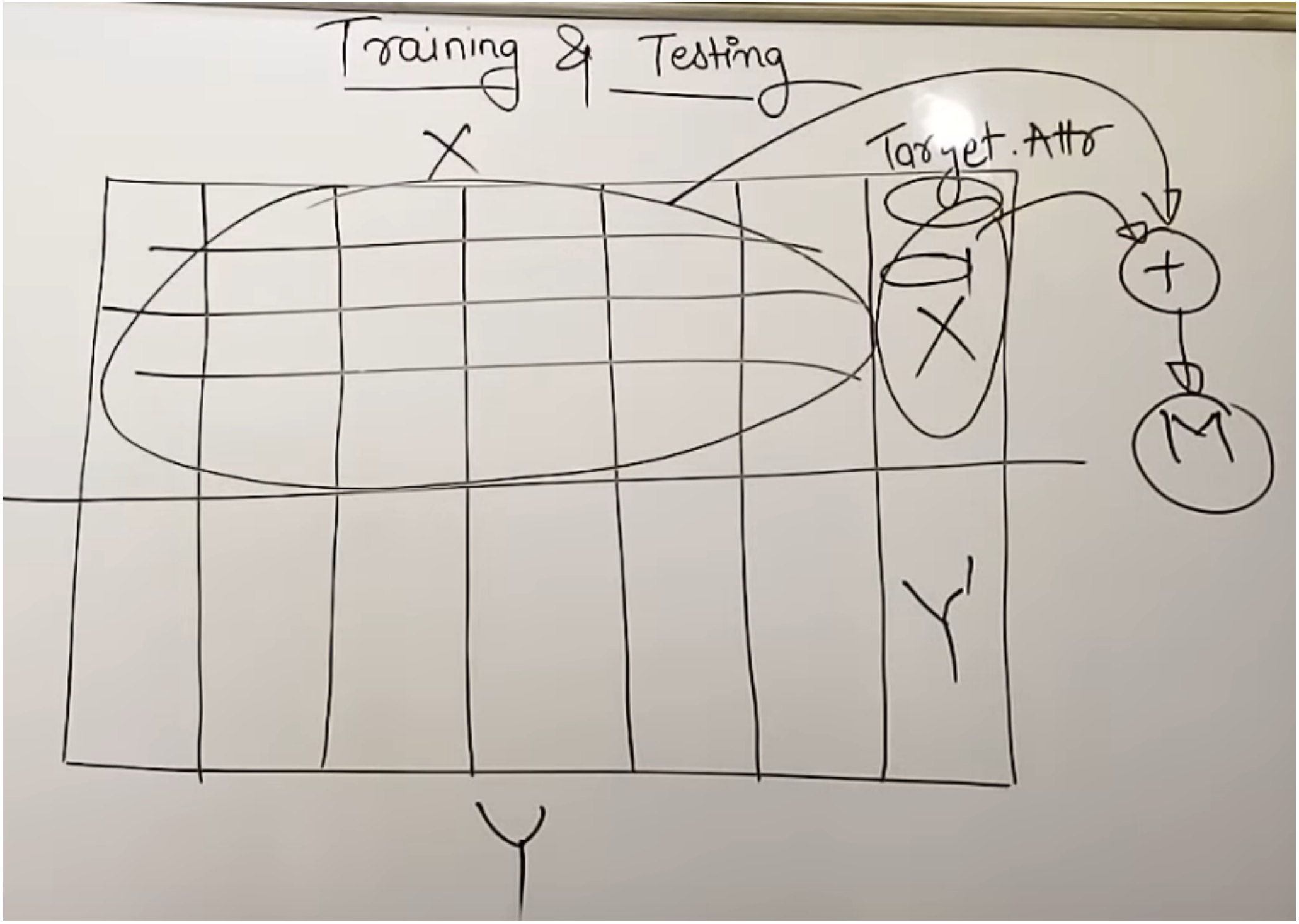
Training & Testing

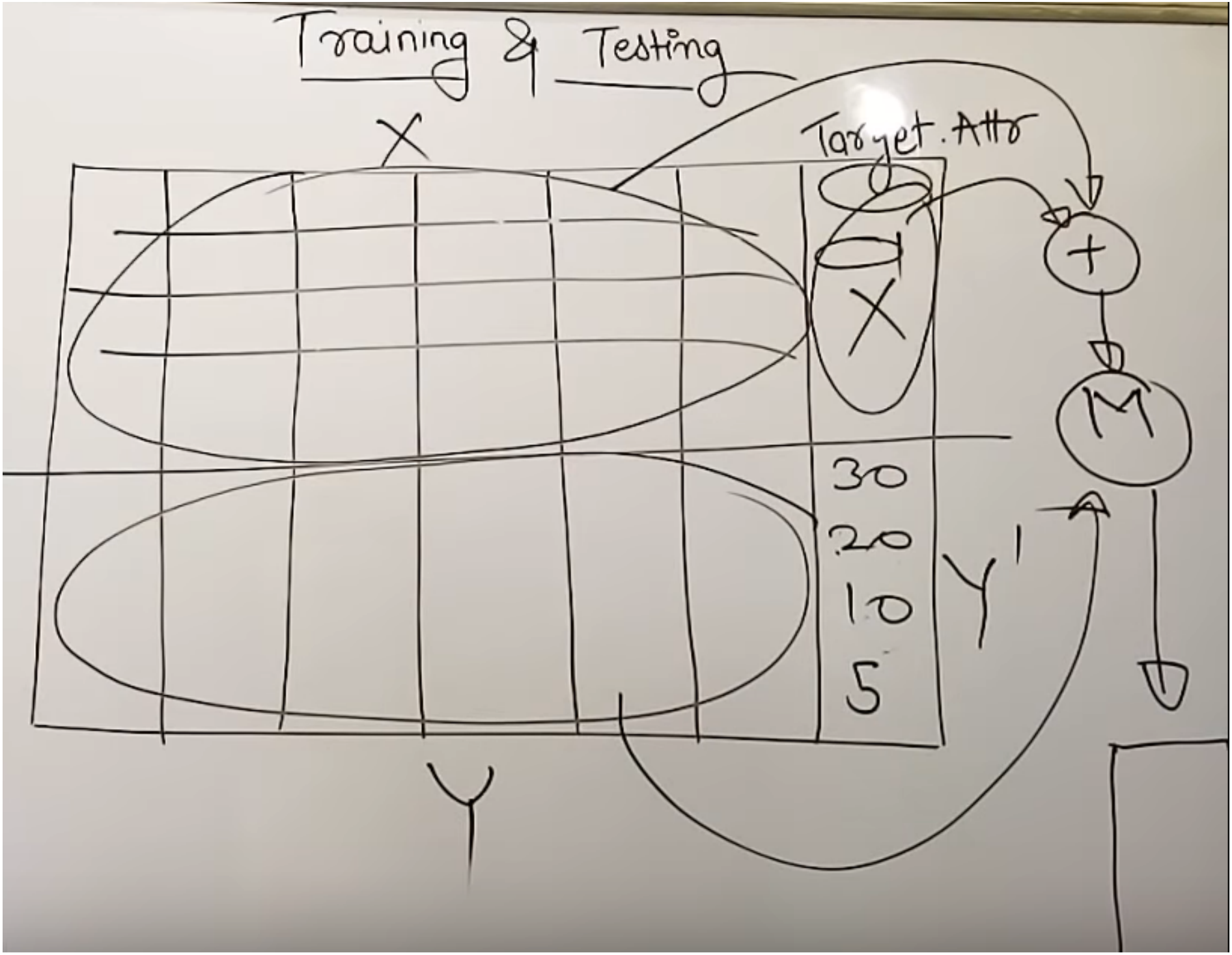


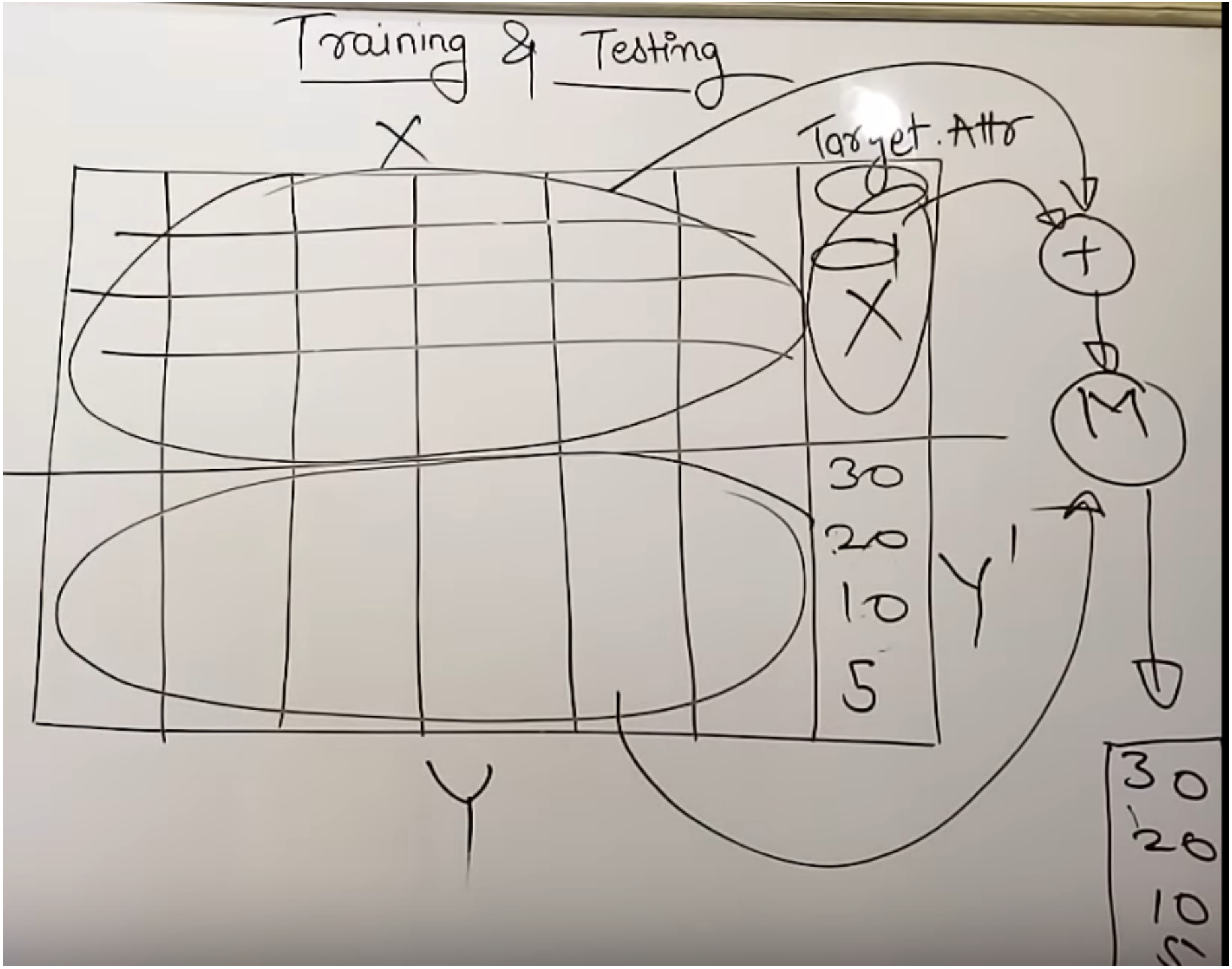

Training & Testing
X
Target. Attr
X'
Y'
Y

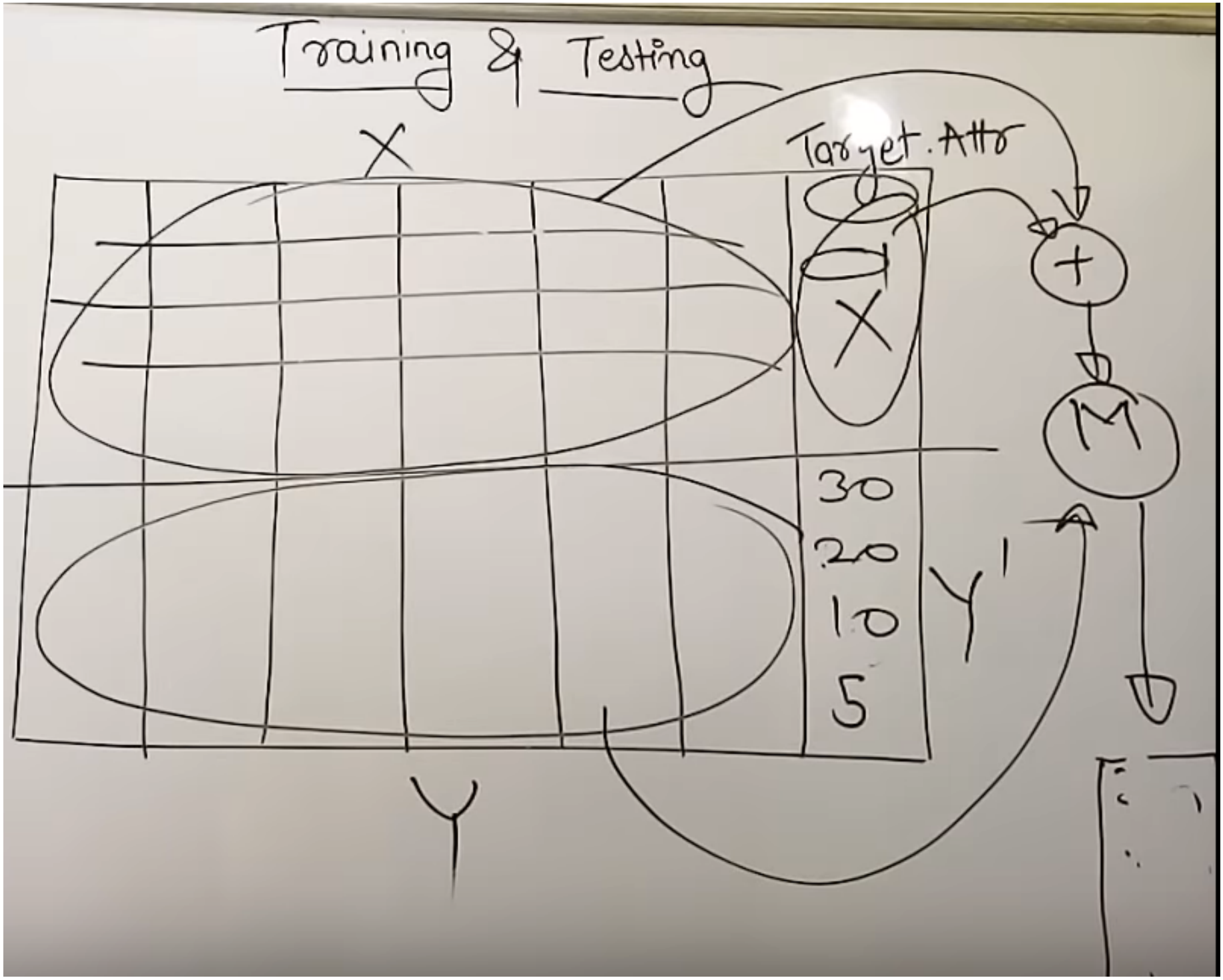

Training & Testing
X
Target Attr
X'
+
Y'
Y


Training & Testing
X
Target. Attr
X'
+
M
Y'
Y


Training & Testing
X
Target. Attr
X
+
M
Y'
Y


Training & Testing
X
Target Attr
30
20
10
5
Y
+
M
Y'


Training & Testing
X
Target. Attr
X
30
20
10
5
Y
Y
+
M
30
20
10


Training & Testing
X
Target. Attr
X
30
20
10
5
Y
Y'
+
M

training & testing data.

Features

Definition: *Features are the individual input variables or attributes that the machine learning algorithm uses to make predictions. These are the characteristics or properties of the data that the algorithm analyzes.*

Example: *In a dataset of houses, features could include square footage, number of bedrooms, number of bathrooms, and location.*

Labels (Target Variable or Output)

Definition: *Labels are the output variables that the machine learning algorithm is trying to predict. They represent the target or the outcome of interest.*

Example: *In a dataset of houses, the label could be the sale price.*

Training Data

Definition: *Training data is the portion of the dataset that is used to train or teach the machine learning algorithm. It consists of both features and corresponding labels. The algorithm learns patterns and relationships in the training data.*

Purpose: *The goal is for the algorithm to generalize from the training data and make accurate predictions on new, unseen data.*

Example: *In a dataset of houses, 80% of the data might be used for training the algorithm.*

Testing Data

Definition: *Testing data is a separate portion of the dataset that is not used during the training phase. It is reserved to evaluate the performance of the trained algorithm. Testing data includes features and corresponding labels.*

Purpose: *The testing data helps assess how well the algorithm generalizes to new, unseen data and whether it can make accurate predictions.*

Example: *In a dataset of houses, 20% of the data might be set aside for testing the algorithm.*

Workflow

Data Splitting:

- *The dataset is typically split into a training set and a testing set. The training set is used for training the model, and the testing set is used for evaluating its performance.*
- *Common split ratios are 80/20 or 70/30, depending on the size of the dataset.*

Training:

- *The algorithm is trained on the training data by learning patterns and relationships between features and labels.*

Testing/Evaluation:

- *The trained model is then tested on the separate testing data to assess its performance. This involves making predictions on the testing features and comparing them to the actual labels.*

Performance Metrics:

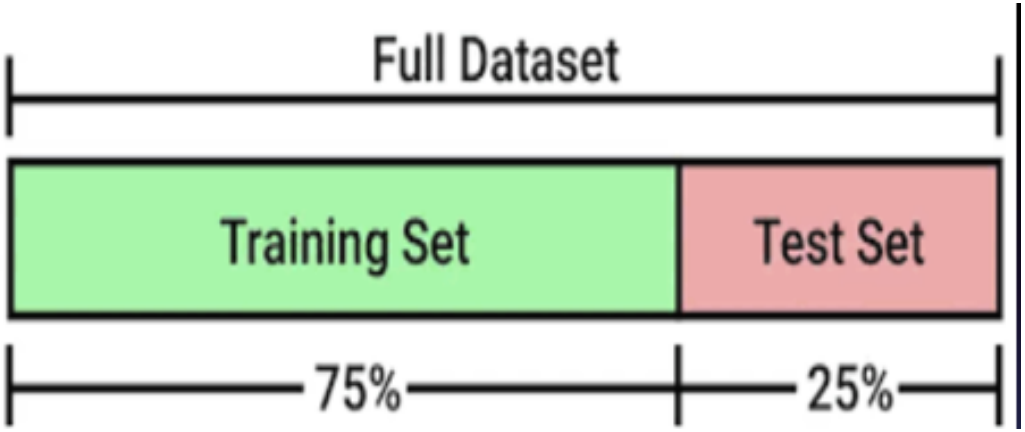
- *Metrics such as accuracy, precision, recall, and F1-score are often used to quantify the performance of the model on the testing data.*

Training a Model

Let say we have sale price, sq ft., year built, number of rooms, tax information of 100 houses.




Training a Model
Let say we have sale price, sq ft., year built, number of rooms, tax information of 100 houses.
Initial Data
=
Training Data
+
Test Data
100 houses
75 houses
25 houses


Full Dataset
Training Set
Test Set
75%
25%

Train-Test Split

Split your dataset into two parts: a training set used to train the model and a test set used to evaluate its performance.

```
from sklearn.model_selection import
train_test_split

X_train, X_test, y_train, y_test =
train_test_split(X, y, test_size=0.2,
random_state=42)
```